DEPTH FIRST SEARCH

AIM:

To explore all vertices of a graph or tree by traversing as far as possible along each branch before backtracking.

```python
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()
    visited.add(start)
    print(start, end=' ')
    for neighbor in graph.get(start, []):
        if neighbor not in visited:
            dfs(graph, neighbor, visited)

# User input
graph = {}
nodes = int(input("Enter number of nodes: "))
for _ in range(nodes):
    node = input("Enter node name: ")
    neighbors = input(f"Enter neighbors of {node} (space-separated): ").split()
    graph[node] = neighbors

start_node = input("Enter start node for DFS: ")
print("DFS Traversal:")
dfs(graph, start_node)
```

RESULT:

A traversal order of nodes (or discovery of a path, component, or solution) depending on the problem, such as a list of visited nodes or detection of cycles, connected components, or paths.